CNN Architectures for Image Classification

MNIST, CIFAR-10 & a channel-attention residual variant

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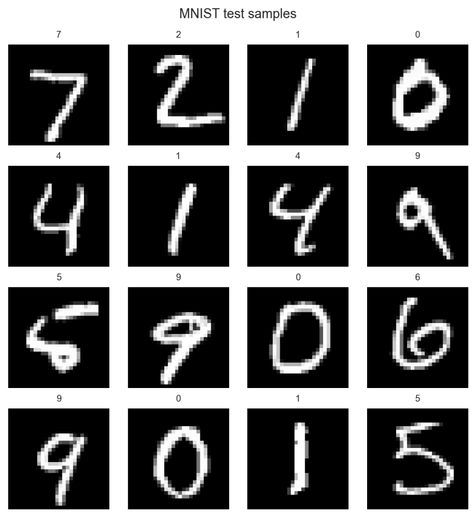
Problem, datasets, and goal

Task. 10-class image classification.

Datasets.

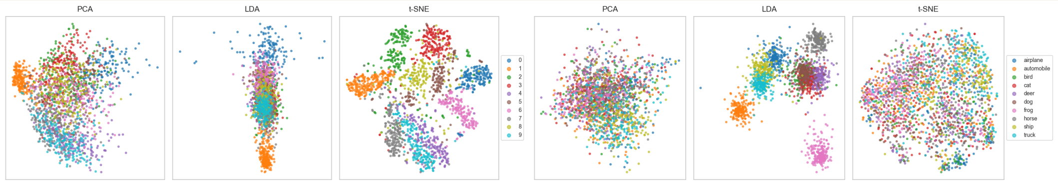
- **MNIST** – handwritten digits, 28×28 grayscale, 60k train / 10k test, 10 classes (0–9).
- **CIFAR-10** – natural objects, 32×32 RGB, 50k train / 10k test, 10 classes (airplane, car, bird, cat, deer, dog, frog, horse, ship, truck).

Goal. Compare seven CNN architectures (one is our own modification of ResNet-18) under a single training pipeline using accuracy, macro precision/recall/F1, and AUROC.



Class structure: PCA → LDA → t-SNE

MNIST raw pixels **CIFAR-10 raw pixels**



Takeaway. MNIST classes form clear clusters even in raw pixel space (LDA separates all 10 digits in 2D) — this explains why even a flat MLP scores 99%. CIFAR-10 is the opposite: all projections collapse into one overlapping cloud. Every percentage point above 50% on CIFAR-10 requires spatial feature learning that only convolution can provide.

Models compared, and our modification

Seven architectures, identical training recipe.

Model	Family
MLP	baseline, no spatial bias
SimpleCNN	2 conv + 2 FC
LeNet-5	classic 1998 CNN
AlexNet*	5 conv + 3 FC
VGG-11*	deep 3×3 + BatchNorm
ResNet-18*	residual, 4 stages
SE-ResNet-Lite	**ours**

* Adapted for 32×32 inputs.

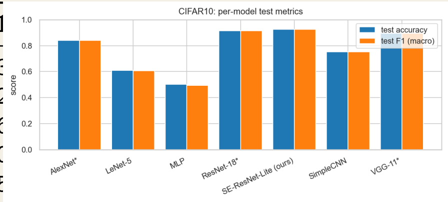
Our novel architecture — SE-ResNet-Lite.

- Take ResNet-18 as the backbone.
- Insert a **Squeeze-and-Excitation block** inside every residual block. SE = global-pool $\rightarrow$ 2-layer MLP $\rightarrow$ sigmoid $\rightarrow$ per-channel gating.
- Replace the stem with a **depthwise-separable** 3×3 DW $\rightarrow$ 1×1 PW.
- Net parameter overhead vs ResNet-18: $<1\%$.

Hypothesis. Channel attention boosts accuracy; the lighter stem keeps the parameter budget essentially fixed.

Results and conclusions

Model	MNIST acc	CIFAR-10 acc	F1
MLP	99.03%	50.18%	0.493
SimpleCNN	99.46%	75.34%	0.752
LeNet-5	99.31%	61.02%	0.606
AlexNet	99.57%	84.03%	0.840
VGG-11	99.62%	89.48%	0.895
ResNet-18	99.65%	91.38%	0.914
SE-ResNet-Lite	**99.70%**	**92.67%**	**0.927**



Limitation. Single random seed — 1–2 pp differences need multiple runs to be statistically meaningful.

Future work. Ablate SE-only vs DW-stem-only; sweep reduction ratio *r*; test on Tiny ImageNet.

Key findings.

- Architecture barely matters on MNIST (0.67 pp spread); it matters enormously on CIFAR-10 (42.5 pp spread, MLP to SE-ResNet-Lite).
- Residual connections outperform VGG-11 by 1.9 pp at similar parameter count — skip connections are doing real work, not just adding depth.